ĐẠI HỌC TÔN ĐỨC THẮNG
TON DUC THANG UNIVERSITY

503043

# Introduction to Artificial Intelligence

## Lecture 2: Intelligent Agents

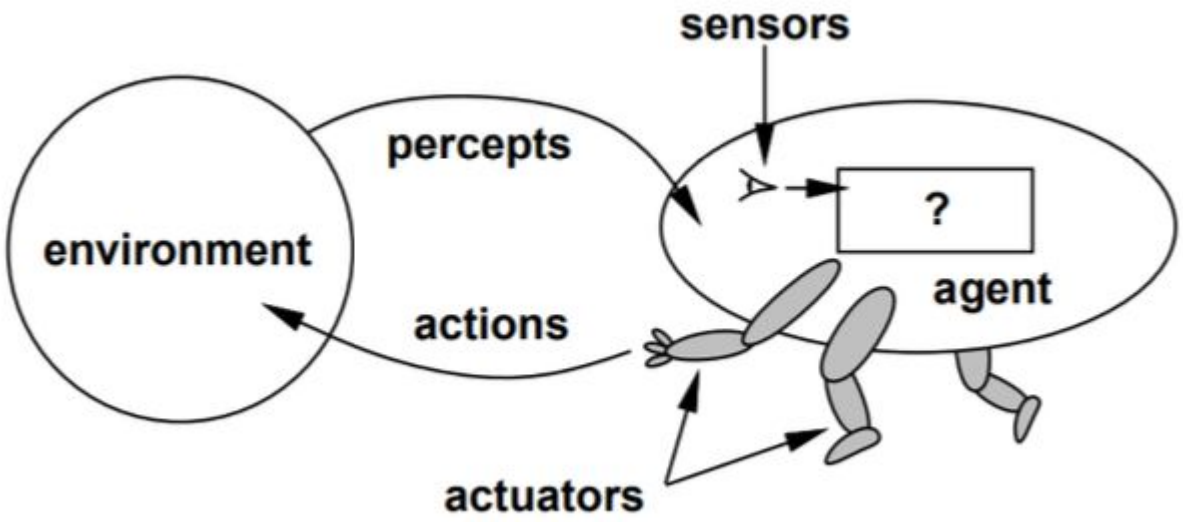
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# Outline

- Agents and Environments
- The Concept of Rationality
- The Nature of Environments
- The Structures of Agents

# What is agent?

- AI studies how to make computers do things that people are better at if they could.
- Such systems are called Agents.
- An agent **perceives** its environment through **sensors** and **acts** upon that environment through **actuators.**



# Examples of Agents

| Agent | Sensors | Actuators |
|---|---|---|
| Human agent | eyes, ears, and other organs. | hands, legs, vocal tract, etc. |
| Robotic agent | cameras, infrared range finders, etc. | levels, motors, etc. |
| Software agent | keystrokes, file contents, network packets, etc. | displaying on screen, writing files, sending network packets, etc. |

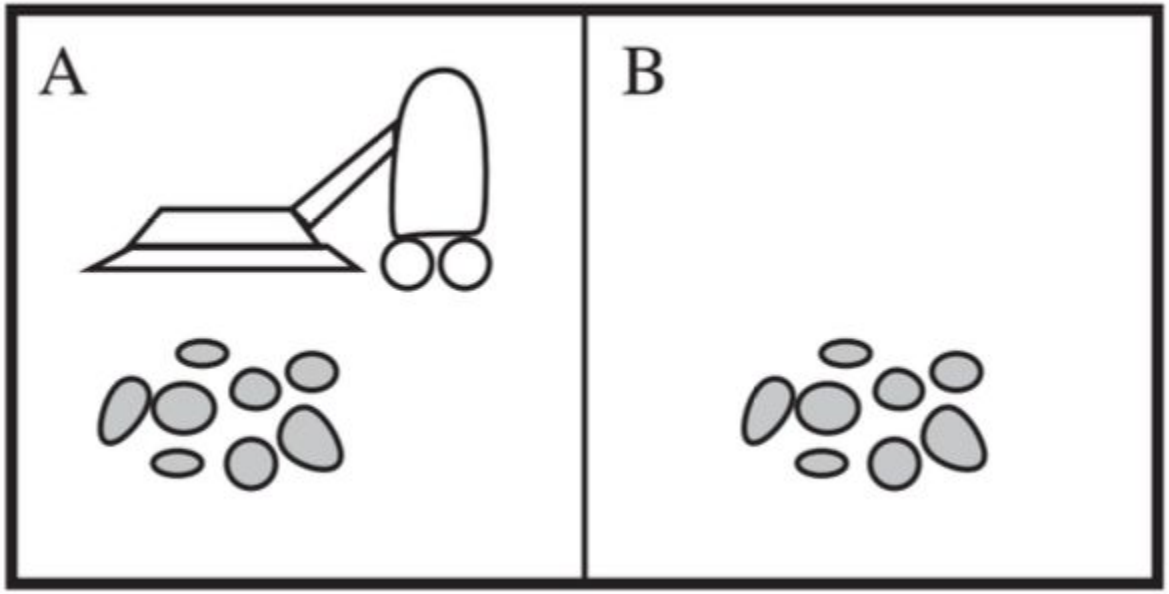
# Agent's Behavior

- Percept: the agent's perceptual inputs at any given instant
- Percept sequence: the complete history of everything the agent has ever perceived
- An agent's behavior is described by the agent function that maps any given percept sequence to an action.

$$f: P^* \rightarrow A$$

# Example: Vacuum Agent

- Percepts: location and contents, e.g., [A,Dirty]
- Actions: Left, Right, Suck, Do Nothing



# Vacuum Agent

| Percept sequence | Action |
|---|---|
| [A, Clean] | Right |
| [A, Dirty] | Suck |
| [B, Clean] | Left |
| [B, Dirty] | Suck |
| … | … |
| [A, Clean], [A, Clean], [A, Clean] | Right |
| [A, Clean], [A, Clean], [A, Dirty] | Suck |

# Vacuum Agent

```
function VACUUM-AGENT([location,status])

returns an action

    if status = Dirty then return Suck

    else if location = A then return Right

    else if location = B then return Left
```

# Importance of agents

- A tool for analyzing systems
- All areas of engineering can be seen as designing artifacts that interact with the world.
- AI designs artifacts that have significant computational resources and the task environment requires nontrivial decision making.

# Rational agents

- A rational agent is one that does the right thing.
- We need ways to measure success → Performance measure.
- Performance measure evaluates any given sequence of environment states.
- General rule: Design performance measures according to
  - What one actually wants in the environment
  - Not how one thinks the agent should behave
- Example: vacuum agent
  - The amount of dirt cleaned up in a single eight-hour shift
  - The floor clean, no matter how the agent behaves

# Rationality

- Performance measure
  - Define the criterion of success
- Prior knowledge
  - What the agent knows about the environment
- Percept sequence
  - The agent's percept to date
- Actions
  - What the agent can perform

# Definition of rational agents

- For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

# Example: Vacuum machine

- Performance measure
  - Award one point for each clean square at each time step, over 10000 timesteps
- Prior knowledge about the environment
  - The geography of the environment (2 squares)
  - The effect of the actions
- Actions that can perform
  - Left, Right, Suck, and Do Nothing
- Percept sequences
  - Where is the agent?
  - Whether the location contains dirt?
- The agent is rational.

# Omniscience, Learning, Autonomy

| | |
|---|---|
| Omniscience | • Knows the actual outcome of its actions in advance<br>• No other possible outcomes<br>• Impossible in real world |
| Learning | • A rational agent must learn as much as possible from what it perceives |
| Autonomy | • A rational agent should be autonomous – Learn what it can to compensate for partial or incorrect prior knowledge. |

# Examples

- Hide and seek
- Apple collector
- https://youtu.be/shccS5kJvtQ
- https://youtu.be/FIz8ycRCSw0

# Task environment

- Task environments are essentially the “problems” to which rational agents are the “solutions”.
- PEAS:
  - **P**erformance measure
  - **E**nvironment
  - Agent’s **A**ctuators
  - Agent’s **S**ensors
- In designing an agent, the first step must always be to specify the task environment (PEAS) as fully as possible.

# Example: Automated taxi driver

- Agent type: taxi driver
- Performance measure: safe, fast, legal, comfortable trip, maximize profits
- Environment: roads, other traffic, pedestrians, passengers.
- Sensors: cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, etc.
- Actuators: steering, accelerator, brake, signal, horn, display.

# Properties of Task Environment

| | |
|---|---|
| Fully observable | Partially observable |
| Single agent | Multiagent |
| Deterministic | Stochastic |
| Episodic | Sequential |
| Static | Dynamic |
| Discrete | Continuous |
| Known | Unknown |

# Properties of Task Environment

- Fully observable: The agent's sensory gives it access to the complete state of the environment.
  - The agent need not maintain internal state to keep track of the world.
- Partially observable
  - Noisy and inaccurate sensors
  - Parts of the state are simply missing from the sensor data, e.g., a vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares
- Unobservable: The agent has no sensors at all

# Properties of Task Environment

- Single agent: An agent operates by itself in an environment.
  - E.g., solving crossword → single-agent,
  - playing chess → two-agent
- Competitive vs. Cooperative multiagent environment
  - E.g., playing chess → competitive,
  - driving on road → cooperative

# Properties of Task Environment

- Deterministic: The next state of the environment is completely determined by the current state and the action executed by the agent.
  - E.g., the vacuum world → deterministic,
  - driving on road → stochastic
- Most real situations are so complex that they must be treated as stochastic.

# Properties of Task Environment

- Episodic: The agent's experience is divided into atomic episodes, in each of which the agent receives a percept and then performs a single action.
  - Quality of action depends just on the episode itself
  - Do not need to think ahead
- Sequential: A current decision could affect future decisions.
  - E.g., spotting defective parts on an assembly line vs. playing chess

# Properties of Task Environment

- Static: The environment is unchanged while an agent is deliberating.
  - E.g., crossword puzzles → static, taxi driving → dynamic
- Dynamic: The agent is continuous asked what it wants to do
  - If it has not decided yet, that counts as deciding to do nothing.
- Semi-dynamic: The environment itself does not change with the passage of time but the agent's performance score does
  - E.g., chess playing with a clock

# Properties of Task Environment

- Discrete vs. continuous
  - The distinction applies to the state of the environment, to the way time is handled, and to the agent's percepts and actions
  - E.g., the chess has a finite number of distinct states, percepts and actions; while the vehicles' speeds and locations sweep through a range of continuous values smoothly over time.
- Known vs. unknown
  - Known environment: the outcomes (or outcome probabilities if the environment is stochastic) for all actions are given.
  - Unknown environment: the agent needs to learn how it works to make good decisions.

# Structures of Agents

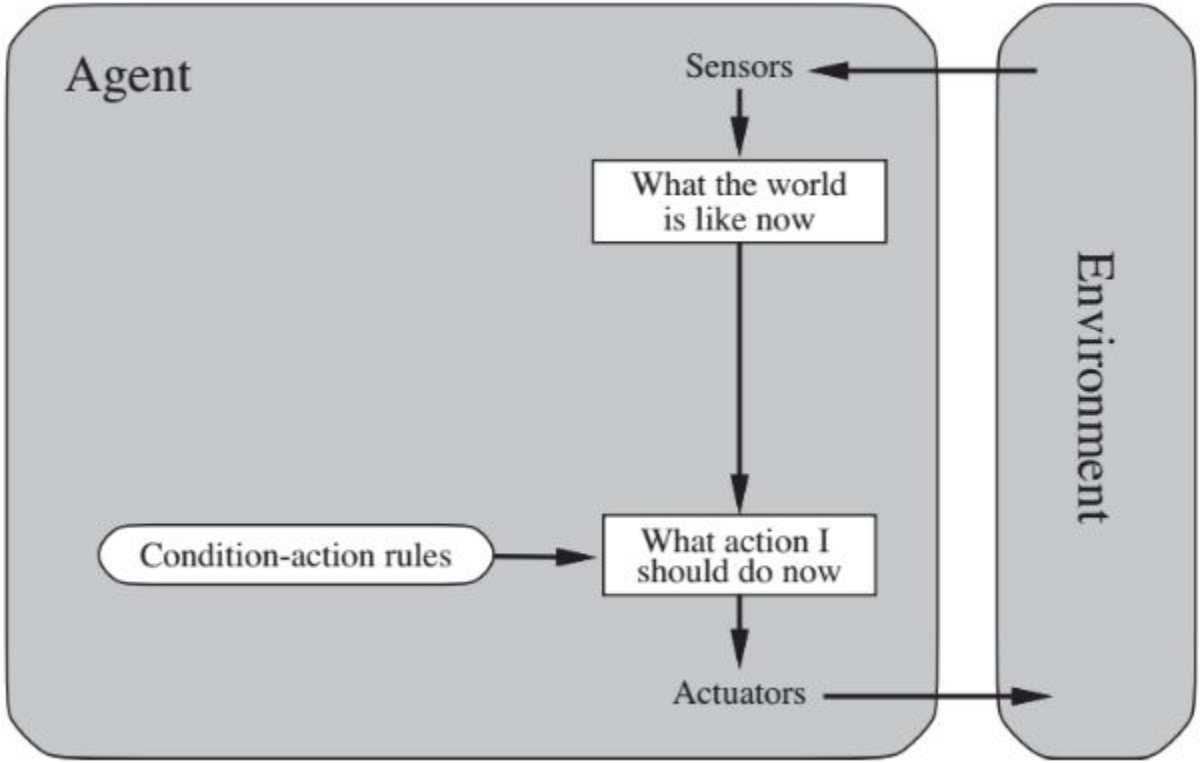
- Simple reflex agents
- Model-based reflex agents
- Goal-based agents
- Utility-based agents

# Simple reflex agents

- Select actions based on the current percept, ignoring the rest of the percept history
- The connection from percept to action is represented by condition-action rules.

**IF** current percept **THEN** action

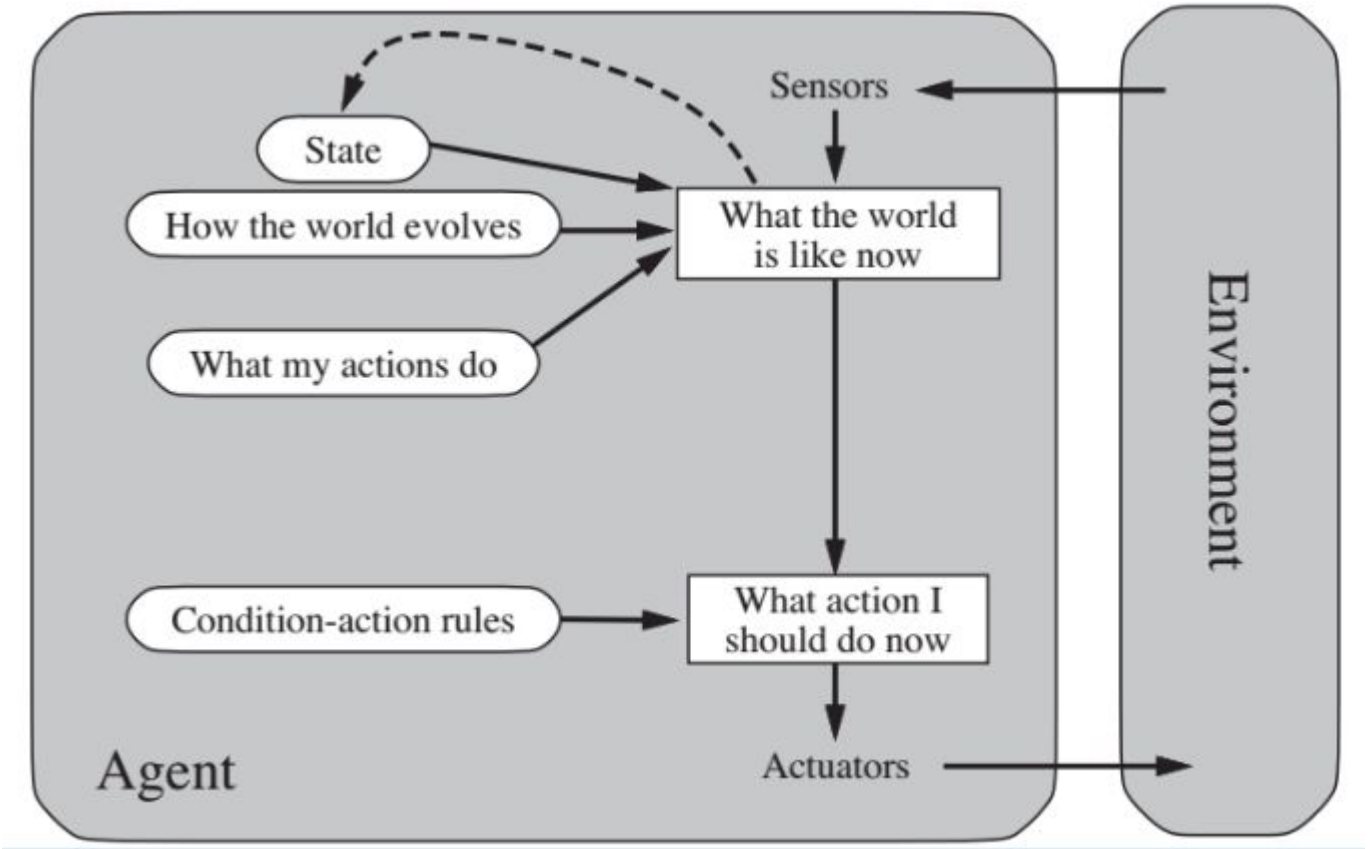
# Simple reflex agents



# Model-based reflex agents

- Partially observable → the agent must keep track of an internal state
  - It depends on the percept history and reflects some of the unobserved aspects
  - E.g., driving a car and changing lane
- The agent program updates the internal state information as time goes by encoding two kinds of knowledge
  - How the world evolves independently of the agent
  - How the agent's actions affect the world

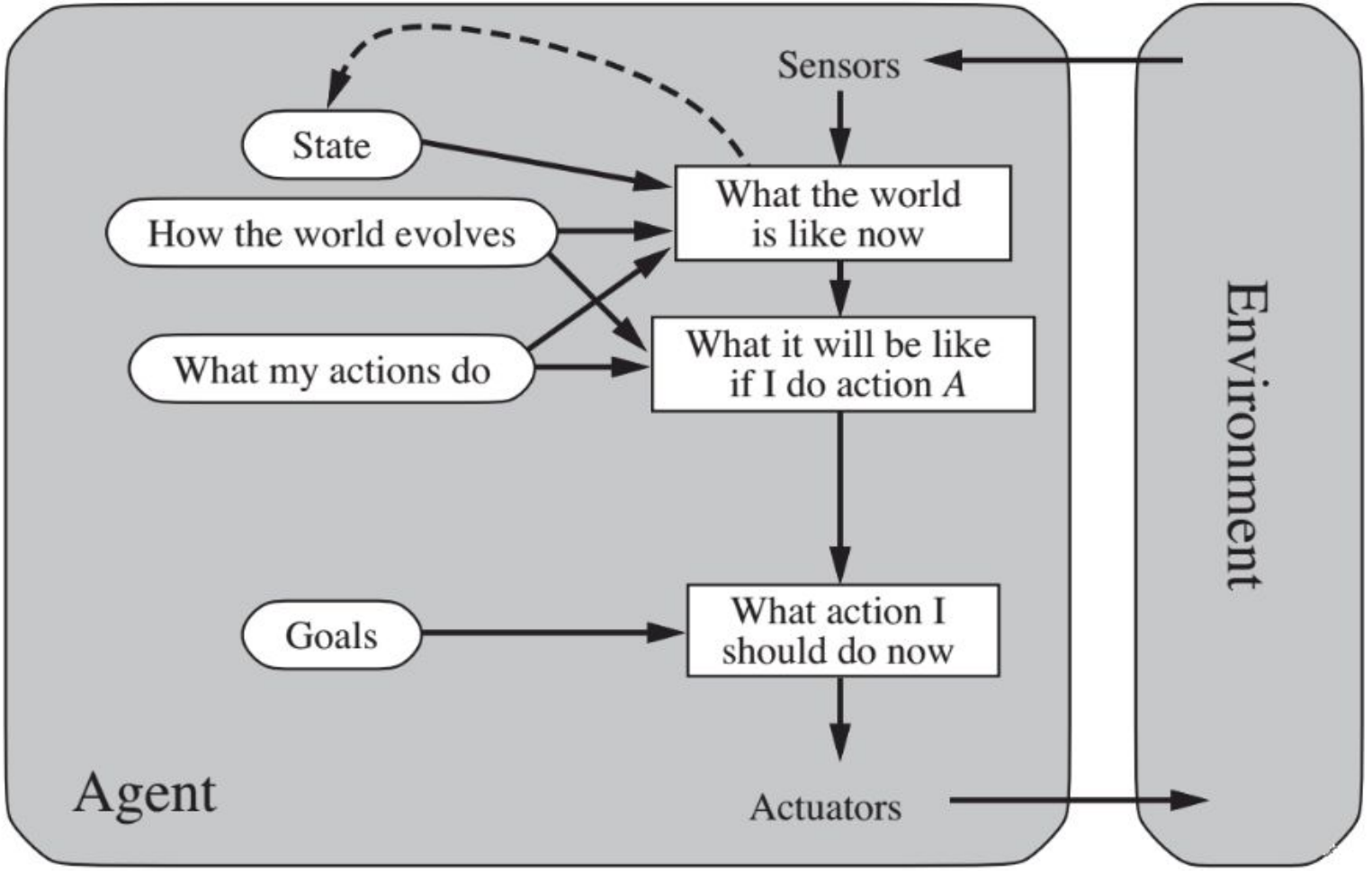
# Model-based reflex agents



# Goal-based agents

- Current state of the environment is always not enough
- The agent further needs some sort of goal information that describes desired situations.
  - E.g., at a road junction, the taxi can turn left, turn right, or go straight on, depending on where the taxi is trying to get to.
- Less efficient but more flexible
  - Knowledge that supports the decisions is represented explicitly and can be modified

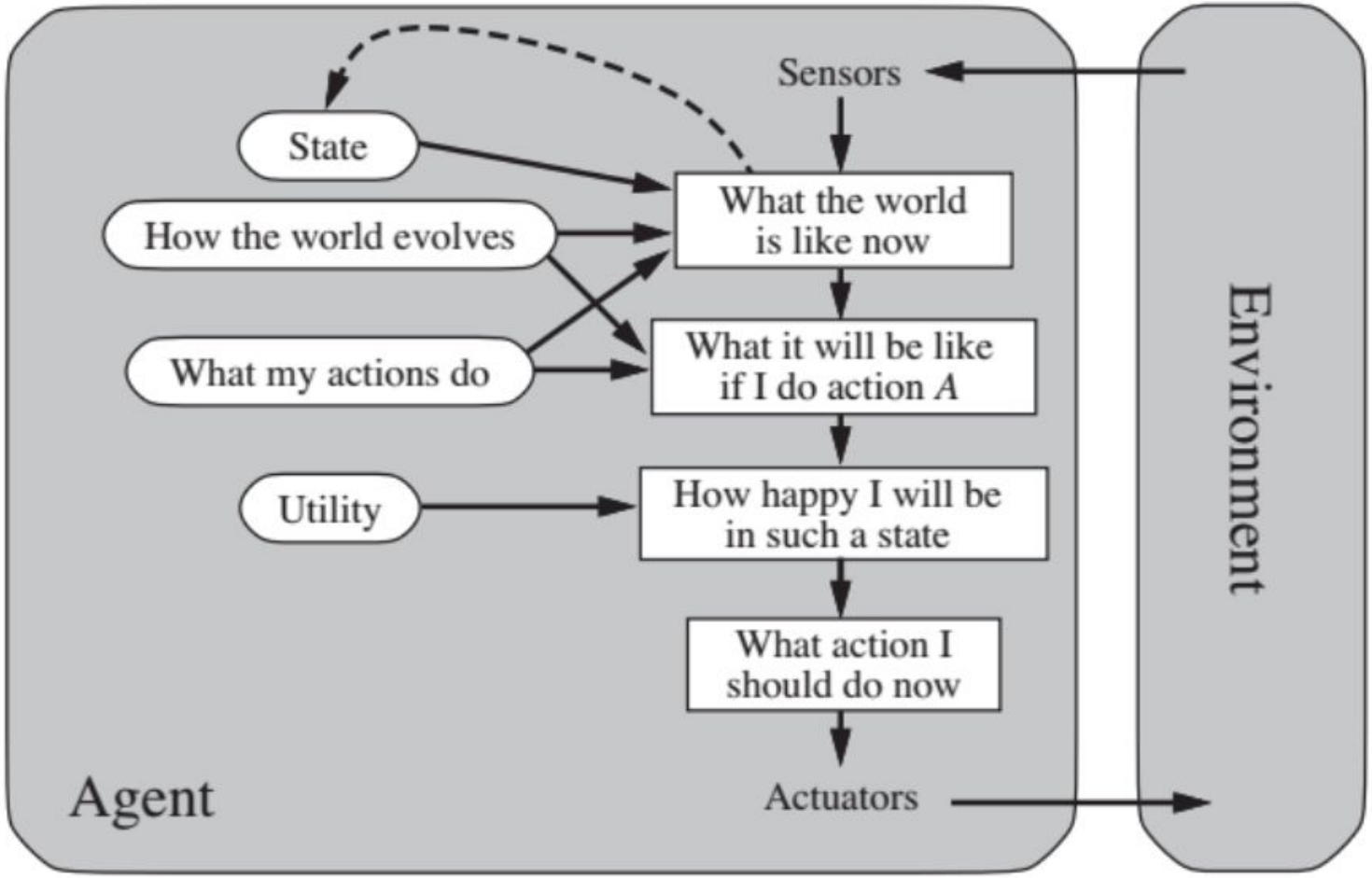
# Goal-based agents



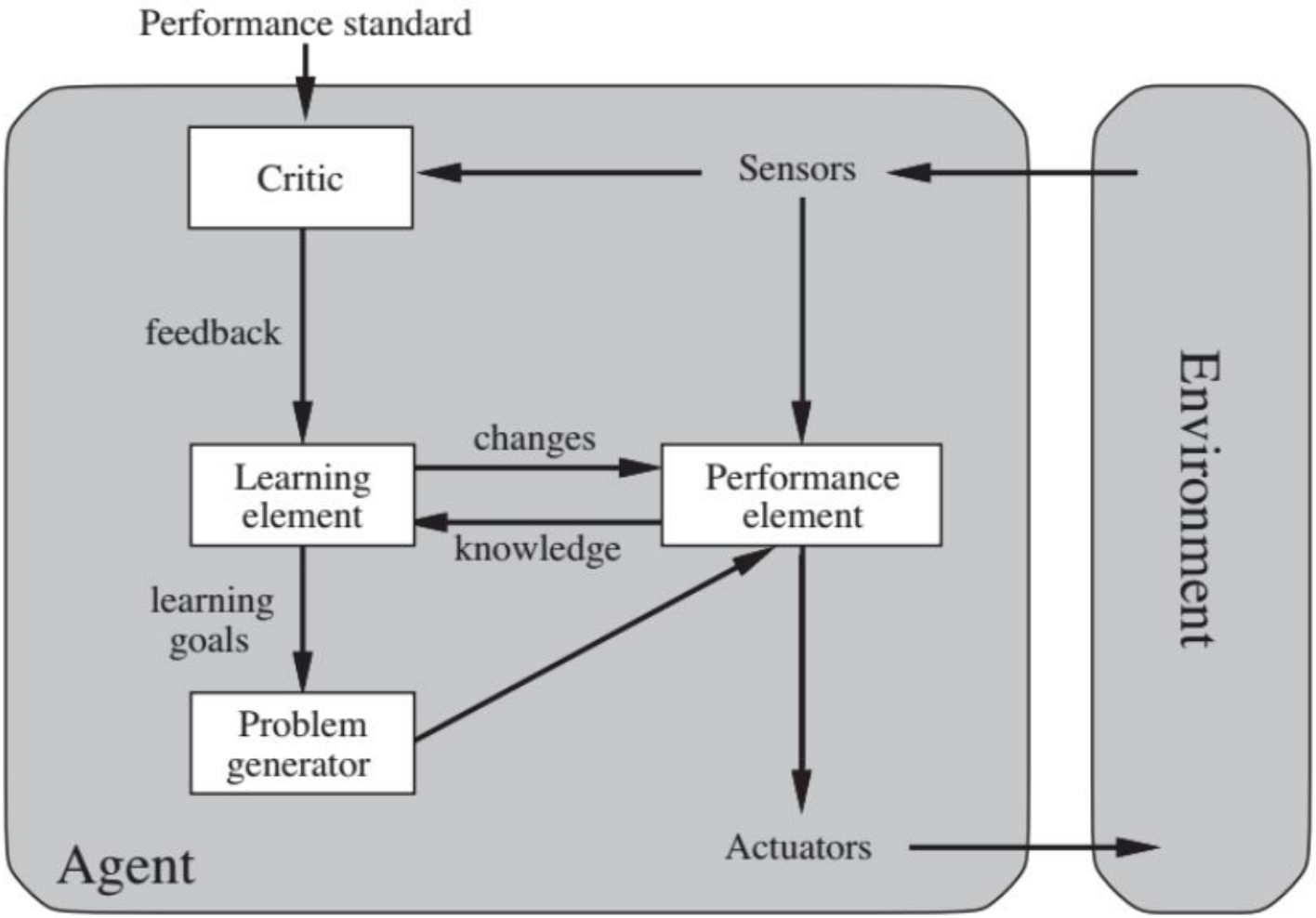
# Utility-based agents

- Goals alone are inadequate to generate high-quality behavior in most environments
  - Many action sequences can get the goals, some are better, and some are worse
  - E.g., go home: Vinasun taxi or Grab car?
- An agent's utility function is essentially an internalization of the performance measure.
  - Goal → success, utility → degree of success (how successful it is)
  - If state A is more preferred than others, then A has higher utility

# Utility-based agents



# Learning Agents



# Learning Agents

- Learning in intelligent agents is a process of modification of each component of the agent to bring the components into closer agreement with the available feedback information, thereby improving the overall performance of the agent.
  - Learning element → Making improvement
  - Performance element → Selecting external actions
  - Critic → Tells the Learning element how well the agent is doing with respect to fixed performance standard. (Feedback from user or examples, good or not?)
  - Problem generator → Suggest actions that will lead to new and informative experiences.

# Homework

- Conduct homework in the given notebook

# References


- Stuart Russell and Peter Norvig. 2009. Artificial Intelligence: A Modern Approach (3rd ed.). Prentice Hall Press, Upper Saddle River, NJ, USA.
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- Nguyễn Ngọc Thảo, Nguyễn Hải Minh. 2020. Bài giảng Cơ sở Trí tuệ Nhân tạo. Khoa Công nghệ Thông tin. Trường ĐH Khoa học Tự nhiên, ĐHQG-HCM.